

# **Estimating spatial firm productivity spillovers using UK microdata**

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## DATA

I have used data from Fame for this investigation. Fame is a comprehensive set of financial datapoints from UK firms, with instruments appropriate for this analysis, in particular number of employees and balance sheet data from which economically interesting variables can be derived. Fame is provided by Moody's Analytics, an international credit agency, and collects its data from UK firms via survey<sup>3</sup>.

Its international counterpart, Orbis, offers a similar service for international counterparts. This analysis could therefore be extended to offer insights comparing the spillover effects of UK firms to their international counterparts. For this investigation, the UK dataset is comprehensive and the most policy-relevant for this investigation.

Raw data was accessed from Fame in February 2025 by Professor Lars Nesheim at UCL for (Irastorza-Fadrique et al., 2026). This project stored a series of Excel tables splitting financial balance sheet variables by their category. Downloadable indicators under 'a1\_ID' and 'a5\_misc' contain fixed datapoints about each firm; 'a2\_key\_finance', 'a3\_assets', and 'a4\_profits' contain yearly financial data about each firm. I designed a series of Python scripts to ingest this data and store it in a 'DuckDB' database, separated into fixed and yearly datapoint tables. 'DuckDB' is an open-source, local, relational (SQL) architecture, designed for processing bulk queries across columns of variables, making it well-suited for econometric data analysis.

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<sup>3</sup>Moody's Analytics (2026)

## MODEL

I start by estimating a basic 2-factor Solow production model at the firm level.

Labour input is taken from the number of employees reported on the FAME balance sheet per firm. This dataset, like all others, does not include hours worked per employee on a firm-level that would offer a more granular productivity (GVA per hour worked) measures. An estimation in this setting would likely have to impute the measure from industry aggregates, which would likely confound the true firm-level signal.

Capital is FAME's `fixed_total` variable, which represents the sum of tangibles, intangibles, and other investments within the balance sheet, namely an extremely comprehensive measure of the firm's fixed capital. The average capital-output ratio is 6x within FAME when filtering for positive values, which is moderately higher than macroeconomic aggregates typically suggest likely due to accounting standard differences.

TABLE I  
Capital-output ratios in FAME corpus of firms

|                      | tangibles | fixed_other<br>(tangibles + intangibles + inv._other) | total_assets<br>(fixed_total + current_assets) |
|----------------------|-----------|-------------------------------------------------------|------------------------------------------------|
| Capital-output ratio | 3.53      | 6.00                                                  | 11.56                                          |

## SPATIAL ANALYSIS

TABLE II  
Baseline TFP calculation from 2-factor production

| Parameter            | gva1_ft    | gva2_ft    | k2_ft      | gva1_f     | gva1_t     | gva1_      |
|----------------------|------------|------------|------------|------------|------------|------------|
| $\beta_0$            | 3.201      | 3.445      | 4.840      | 3.223      | 1.994      | 3.201      |
| const                | (0.032)*** | (0.051)*** | (0.019)*** | (0.027)*** | (0.010)*** | (0.032)*** |
| $\beta_1$            | 0.621      | 0.624      | 0.717      | 0.618      | 0.559      | 0.621      |
| ln_employees         | (0.005)*** | (0.007)*** | (0.005)*** | (0.004)*** | (0.002)*** | (0.005)*** |
| $\beta_2$            |            |            | 0.058      |            |            |            |
| ln_fixed_total       |            |            | (0.001)*** |            |            |            |
| $\beta_3$            | 0.265      | 0.245      |            | 0.264      | 0.422      | 0.265      |
| ln_total_assets      | (0.004)*** | (0.007)*** |            | (0.004)*** | (0.002)*** | (0.004)*** |
| Observations         | 1,081,520  | 618,588    | 1,042,854  | 1,081,520  | 1,081,520  | 1,081,520  |
| i                    | 140,458    | 93,041     | 135,428    | 140,458    |            | 140,458    |
| t                    | 19         | 19         | 19         |            | 19         | 19         |
| $R^2(\text{within})$ | 0.386      | 0.323      | 0.339      | 0.386      | 0.366      | 0.386      |

TABLE III  
Static panel linear-in-means using spatial-group TFP averages

| Parameter                | lmm        | lmm_no_firm_fe | lmm_employees |
|--------------------------|------------|----------------|---------------|
| $\beta_0$                | 2.075      | 0.310          | 1.202         |
| const                    | (0.044)*** | (0.038)***     | (0.215)***    |
| $\beta_1$                | 0.033      | 0.204          | 0.306         |
| peer_tfp_ttwa_donut      | (0.013)*** | (0.013)***     | (0.071)***    |
| $\beta_2$                | 0.015      | 0.142          | 0.015         |
| peer_tfp_pc4_donut       | (0.005)*** | (0.008)***     | (0.005)***    |
| $\beta_3$                | 0.280      | 0.559          | 0.280         |
| peer_tfp_pc8             | (0.005)*** | (0.004)***     | (0.005)***    |
| $\beta_4$                |            |                | -0.058        |
| peer_tfp_pc8_x_employees |            |                | (0.015)***    |
| $\beta_5$                |            |                | 0.184         |
| ln_employees             |            |                | (0.044)***    |
| Observations             | 764,537    | 764,537        | 764,537       |
| i                        | 107,143    |                | 107,143       |
| t                        | 19         | 19             | 19            |
| $R^2(within)$            | 0.040      | -0.000         | 0.040         |

## BIBLIOGRAPHY

### REFERENCES

Aitor Irastorza-Fadrique, Lars Nesheim, and Peter Levell. Three Not-so-easy Puzzles : Post Financial Crisis Performance of the UK Corporate Sector. (*forthcoming*), 2026.

Moody's Analytics. Fame, 2026.

TABLE IV  
Static panel using continuous distance-decay weighted TFP average of peers

| Parameter       | dyn1            | dyn2            | dyn3            |
|-----------------|-----------------|-----------------|-----------------|
| $\beta_0$       | 1.371           | 1.577           | 1.064           |
| const           | (0.134)***      | (0.144)***      | (0.103)***      |
| $\beta_1$       | 0.005           | 0.005           | 0.005           |
| L1.ln_gva1      | (0.001)***      | (0.001)***      | (0.001)***      |
| $\beta_2$       | 0.696           | 0.689           | 0.709           |
| ln_employees    | (0.038)***      | (0.037)***      | (0.040)***      |
| $\beta_3$       | 0.259           | 0.266           | 0.247           |
| ln_total_assets | (0.041)***      | (0.041)***      | (0.044)***      |
| $\beta_4$       | 0.503           |                 |                 |
| tfp_wav1        | (0.028)***      |                 |                 |
| $\beta_5$       |                 | 0.423           |                 |
| tfp_wav2        |                 | (0.023)***      |                 |
| $\beta_6$       |                 |                 | 0.618           |
| tfp_wav3        |                 |                 | (0.042)***      |
| Hansen Test     | X<br>(0.417)    | X<br>(0.459)    | X<br>(0.246)    |
| AR(1) Test      | ✓<br>(0.000)*** | ✓<br>(0.000)*** | ✓<br>(0.000)*** |
| AR(2) Test      | X<br>(0.746)    | X<br>(0.600)    | X<br>(0.945)    |
| Observations    | 713,339         | 713,339         | 713,339         |
| i               | 139,162         | 139,162         | 139,162         |

TABLE V  
Dynamic panel using instrumented continuous distance-decay weighted TFP

| Parameter                | lmm        | lmm_no_firm_fe | lmm_employees |
|--------------------------|------------|----------------|---------------|
| $\beta_0$                | 2.075      | 0.310          | 1.202         |
| const                    | (0.044)*** | (0.038)***     | (0.215)***    |
| $\beta_1$                | 0.033      | 0.204          | 0.306         |
| peer_tfp_ttwa_donut      | (0.013)*** | (0.013)***     | (0.071)***    |
| $\beta_2$                | 0.015      | 0.142          | 0.015         |
| peer_tfp_pc4_donut       | (0.005)*** | (0.008)***     | (0.005)***    |
| $\beta_3$                | 0.280      | 0.559          | 0.280         |
| peer_tfp_pc8             | (0.005)*** | (0.004)***     | (0.005)***    |
| $\beta_4$                |            |                | -0.058        |
| peer_tfp_pc8_x_employees |            |                | (0.015)***    |
| $\beta_5$                |            |                | 0.184         |
| ln_employees             |            |                | (0.044)***    |
| Observations             | 764,537    | 764,537        | 764,537       |
| i                        | 107,143    |                | 107,143       |
| t                        | 19         | 19             | 19            |
| $R^2(within)$            | 0.040      | -0.000         | 0.040         |

#### APPENDIX 1

## APPENDIX 2

## APPENDIX 3